Tensor Product $V \otimes W$ $(= \mathcal{X}/\mathcal{Y})$

$\cong$ (不嚴謹)

$(a\underline{v}_1 + b\underline{v}_2) \otimes \underline{w} = a(\underline{v}_1 \otimes \underline{w}) + b(\underline{v}_2 \otimes \underline{w})$

Theorem (Universal Property, not examinable):

$f: V \times W \to U$ is a 2-linear map

$\Downarrow$

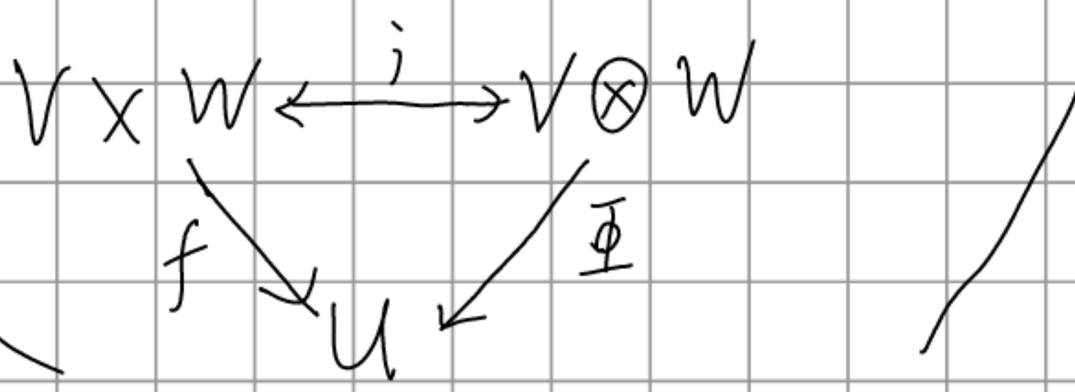
$\exists\, \Phi: V \otimes W \to U$,

s.t. Φ is a linear trans ①

$\Phi(\underline{v} \otimes \underline{w}) = f(\underline{v}, \underline{w})$ ②

(study Φ $\rightarrow\rightarrow$ understand f)

$i(\underline{v}, \underline{w}) := \underline{v} \otimes \underline{w}$ is an injective map



Proof (of Thm.):

Consider $\pi : \mathcal{X} \to U$,

$\mathcal{X} \downarrow = \mathrm{Span}(\mathcal{S})$

$\pi\left(\underbrace{\sum_{i=1}^{n} a_i \{\underline{v}_i, \underline{w}_i\}}_{\in \mathrm{Span}(\mathcal{S})}\right) := \underbrace{\sum_{i=0}^{n} a_i f(\underline{v}_i, \underline{w}_i)}_{\in U}$, $\forall a_i \in \mathbb{R}$

Note: ① π is a l. trans.

② $\{\underline{v}+\underline{v}', \underline{w}\} - \{\underline{v}, \underline{w}\} - \{\underline{v}', \underline{w}\} \in \ker(\pi)$

—— (双线性)

③ —— (双线性)

By ③

and a thm. about quotient space,

we have:

$\pi : \mathcal{X}/\mathcal{Y} \to U$, $\pi[\langle \underline{v}, \underline{w} \rangle + \mathcal{Y}] := \pi(\langle \underline{v}, \underline{w} \rangle)$

$= f(\underline{v}, \underline{w})$

$\downarrow$

$= \Phi$

Theorem 2:

$\underbrace{\qquad\qquad}_{(2^{nd}\text{ last})}$

Proof (not examinable):

(General element in $V \otimes W$ is of the form: $(\underline{v}_1 \otimes \underline{w}_1 + \cdots + \underline{v}_k \otimes \underline{w}_k)$

① ⇓

$\mathcal{C}$ spans $V \otimes W$

② (WTS: $\mathcal{C}$ is L.i.)

Consider $\sum_{j=1}^{m} \sum_{i=1}^{n} \gamma_{i,j} (\underline{v}_i \overset{i}{\otimes} \underline{w}_j) = \underline{0}$ (**)

(WTS: $\forall i, j,\ \gamma_{i,j} = 0$)

dual bases $A^* = \{\phi_1, \cdots, \phi_n\}$ of V^*

$B^* = \{\psi_1, \cdots, \psi_m\}$ of W^*

Let $\varepsilon(\underline{v}, \underline{w}) := \phi_i(\underline{v})\,\psi_j(\underline{w})$

(E.g.: $\varepsilon(k\underline{v}, \underline{w}) = \phi_i(k\underline{v}_i)\,\psi_j(\underline{w})$

$\overset{\text{(*)}}{=} k\,\phi_i(\underline{v}_i)\,\psi_j(\underline{w})$

$= k\,\varepsilon(\underline{v}, \underline{w})$)

By universal property,

$\exists$ l.trans $E: V \otimes W \to \mathbb{F}$,

s.t. $E(\underline{v} \otimes \underline{w}) = \varepsilon(\underline{v}, \underline{w}) = \phi_i(\underline{v})\,\psi_j(\underline{w})$

Apply E to (**)

↳ (兄弟)

Theorem 3:

$\left.\begin{array}{l} T: V \to V' \\ S: W \to W' \end{array}\right\}$ are linear trans,

$\Downarrow$

$\exists$ a linear trans. $T \otimes S : V \otimes W \to V' \otimes W'$,

satisfying $T \otimes S (v \otimes w) = T(v) \otimes S(w)$

Proof:

Consider $g : V \times W \to V' \otimes W'$,

$g(v, w) := T(v) \otimes S(w)$

$\downarrow$

2-linear map

e.g.: ~~~ $(\mathbb{R}, \cdot)$

E.g.: ($\mathbb{R}^2$, usual)

$$T: \mathbb{R}^2 \to \mathbb{R}^3, \quad T(\underline{z}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \underline{z}$$

$$S: \mathbb{R}^2 \to \mathbb{R}^3, \quad S(\underline{y}) = \left(\begin{array}{c} \sim \\ \sim \\ \sim \end{array}\right) \underline{y}$$

$A = \{\underline{e}_1, \underline{e}_2\}$

$B = \{\underline{f}_1, \underline{f}_2, \underline{f}_3\}$ usual basis of $\mathbb{R}^2$ / $\mathbb{R}^3$

$T \otimes S: \mathbb{R}^2 \otimes \mathbb{R}^3 \to \mathbb{R}^2 \otimes \mathbb{R}^3$

by Thm. 2,

a basis of $\mathbb{R}^2 \otimes \mathbb{R}^3$ is: $\ell = \{e_1 \otimes f_1, \cdots$ (6 ↓)

Let's compute $(T \otimes S)_{\ell\ell}$

$$(T \otimes S)(e_1 \otimes f_1) = T(e_1) \otimes S(f_1) = \begin{pmatrix} a \\ c \end{pmatrix} \otimes \begin{pmatrix} p \\ s \\ v \end{pmatrix}$$

$$= ap(e_1 \otimes f_1) + as(e_1 \otimes f_2) + av(e_1 \otimes f_3) + cp(e_2 \otimes f_1) + cs(e_2 \otimes f_2) + cv(e_2 \otimes f_3)$$

$(T \otimes S)(e_1 \otimes f_2)$ " $(e_1 \otimes f_3)$ " $(e_2 \otimes f_1)$ " $(e_2 \otimes f_2)$ " $(e_2 \otimes f_3)$

$$T_{\ell\ell} = \begin{pmatrix} ap & aq & ar & bp & bq & br \\ as & at & au & bs & bt & bu \\ av & aw & ax & bv & bw & bx \\ cp & cq & cr & dp & dq & dr \\ cs & ct & cu & ds & dt & du \\ cv & cw & cx & dv & dw & dx \end{pmatrix}$$

(1st column, 5th column)

$$T \otimes S(e_2 \otimes f_2) = T(e_2) \otimes S(f_2) = \begin{pmatrix} b \\ d \end{pmatrix} \otimes \begin{pmatrix} q \\ t \\ w \end{pmatrix}$$

$$\equiv \left(\begin{array}{c|c} a S_{BB} & b S_{BB} \\ \hline c S_{BB} & d S_{BB} \end{array}\right) \quad \text{(Kronecker product)}$$

Summary on tensor products:

	$V \otimes W$	$V_1 \otimes V_2 \otimes \dots \otimes V_m$
m-linearity	$v \otimes (a\underline{w} + b\underline{w}') = a(v \otimes \underline{w}) + b(v \otimes \underline{w}')$	$\underline{v}_1 \otimes \dots \otimes (a\underline{v}_i + b\underline{v}_i') \otimes \dots \otimes \underline{v}_m = a(\underline{v}_1 \otimes \dots \otimes \underline{v}_i \otimes \dots \otimes \underline{v}_m) + b(\underline{v}_1 \otimes \dots \otimes \underline{v}_i' \otimes \dots \otimes \underline{v}_m)$
Universal property	$(f: V \times W \to U$ bilinear$) \rightsquigarrow (\Phi: V \otimes W \to U$ linear trans$)$	$(f: V_1 \times \dots \times V_m \to U$ is m-linear$) \rightsquigarrow (\Phi: V_1 \otimes \dots \otimes V_m \to U$ linear$)$
General element	$\underline{v}^{(1)} \otimes \underline{w}^{(1)} + \dots + \underline{v}^{(k)} \otimes \underline{w}^{(k)}$	$\underline{v}_1^{(1)} \otimes \dots \otimes \underline{v}_m^{(1)} + \dots + \underline{v}_1^{(k)} \otimes \dots \otimes \underline{v}_m^{(k)}$
dimension	$\dim(V) \cdot \dim(W)$	$\dim(V_1) \cdot \dim(V_2) \dots \dim(V_m)$
linear trans.	$(T: V \to V', S: W \to W') \rightsquigarrow (T \otimes S: V \otimes V' \to W \otimes W')$	$\begin{pmatrix} T_1: V_1 \to V_1' \\ \vdots \\ T_m: V_m \to V_m' \end{pmatrix} \rightsquigarrow \begin{pmatrix} T_1 \otimes \dots \otimes T_m: V_1 \otimes \dots \otimes V_m \to V_1' \otimes \dots \otimes V_m' \\ \text{s.t. } T_1 \otimes \dots \otimes T_m(\underline{v}_1 \otimes \dots \otimes \underline{v}_m) = T_1(\underline{v}_1) \otimes \dots \otimes T_m(\underline{v}_m) \end{pmatrix}$